Motif Analysis in Financial and Macroeconomic Time Series

Motif Analysis – Repeated Dominant Shape Analysis

- **Motif analysis** identifies recurring local patterns in time series, not just overall trend or average behaviour
- Financial and macroeconomic data are often **nonlinear, state-dependent, asymmetric, noisy, and regime-driven**
- The same shock or signal can have very different effects in **calm vs stressed** or **expansion vs contraction** states
- **Linear and nonlinear models can learn recurring structure**, **but motif analysis is purpose-built to make that recurrence explicit, comparable, and directly analysable.**
- **Motif analysis** helps uncover **repeated but not identical behaviours**, supporting **regime identification, early warning, contextual forecasting, and risk monitoring**

Motif Analysis

Technical View: Matrix Profile

- A fixed-length sliding window is moved across the series to create overlapping **subsequences**
- Each subsequence is compared with all others to measure similarity and detect repeated shapes
- The **matrix profile** stores, for each subsequence, the distance to its nearest non-trivial match
- **Low matrix-profile values** indicate **motifs**: recurrent patterns;
- **High matrix-profile values** indicate **discords**: unusual or anomalous behaviour
- **In a nutshell**: **Matrix Profile Framework** provides for:
  - Motif discovery
  - Anomaly detection
  - Segmentation
  - Regime analysis
  - Motif-based feature engineering

Motif Analysis vs. HMM

- **Motif analysis is pattern-based:**
  - **Detects recurring local subsequence/patterns in the data**
  - **Focus on self-similarity and recurrence**
  - **Limited parametric assumptions**

- **HMM and regime-switching models are state-based:**
  - Infers **latent regimes / hidden states and estimate probabilities of switching between them**
  - **Focus on the current system state**
  - **Focus on state persistence and switching dynamics**
  - Requires assumptions on **number of states, transitions, and state-dependent distributions**

Motifs describe repeated shapes; regimes describe persistent underlying states

Motif Analysis Scale Issues

Motifs are not inherently scale-invariant; scale invariance is imposed by the similarity measure and preprocessing choice.

With normalization, motifs represent repeated **shapes**; without it, they represent repeated **shape-plus-magnitude** patterns.

Slide-ready version

Repeated shapes are only scale-invariant if the motif definition makes them so. With z-normalization or similar preprocessing, motif analysis emphasizes shape similarity; without normalization, both shape and magnitude affect whether two episodes are treated as the same motif.

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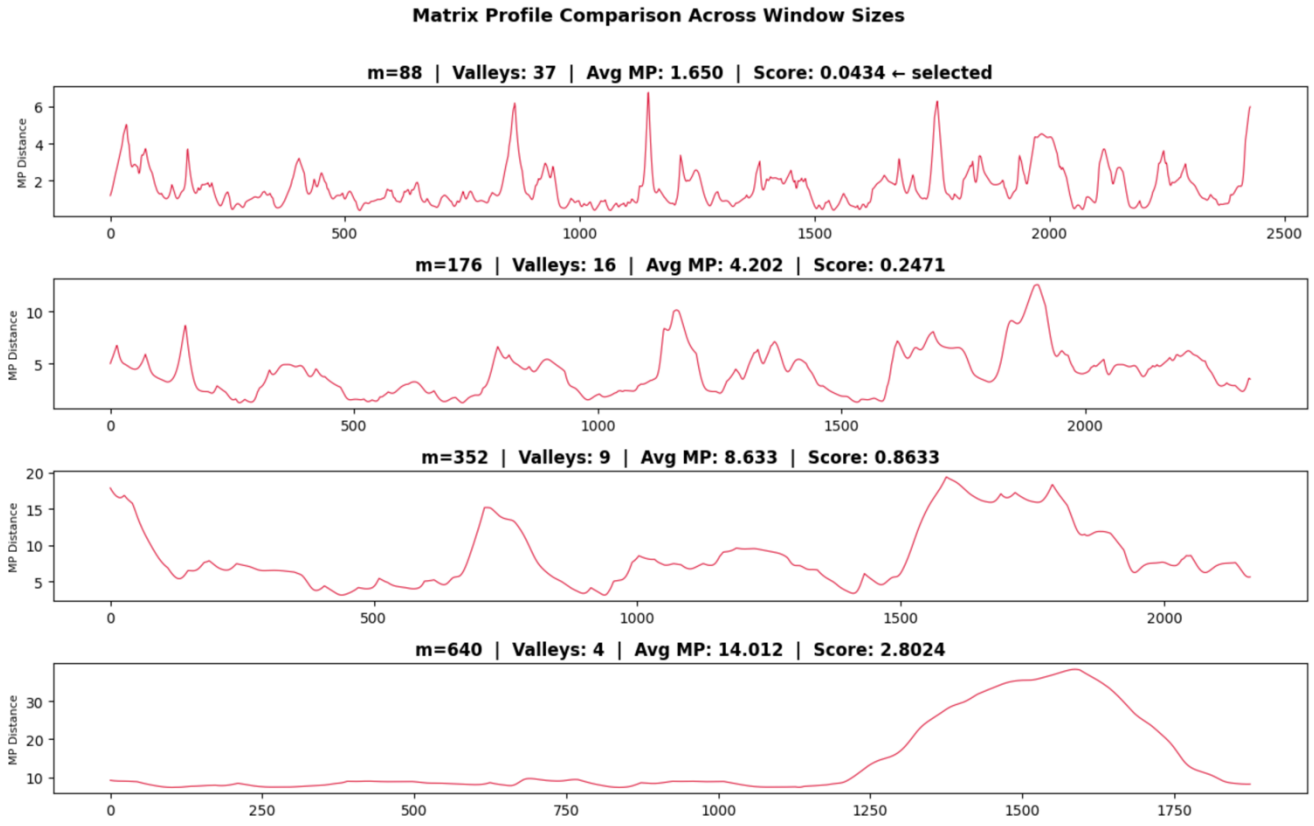
Motif Analysis - Data

- **Ticker** = “SPGI”
- **Data Frequency** = Daily
- HP_LAMB = 1600
- **Data vector** - (_, **trend** = sm.tsa.filters.hpfilter(df['Close'], lamb=HP_LAMB))
  - HP Filter - separates the series into a slow-moving trend component and a cyclical/noise component

Core question

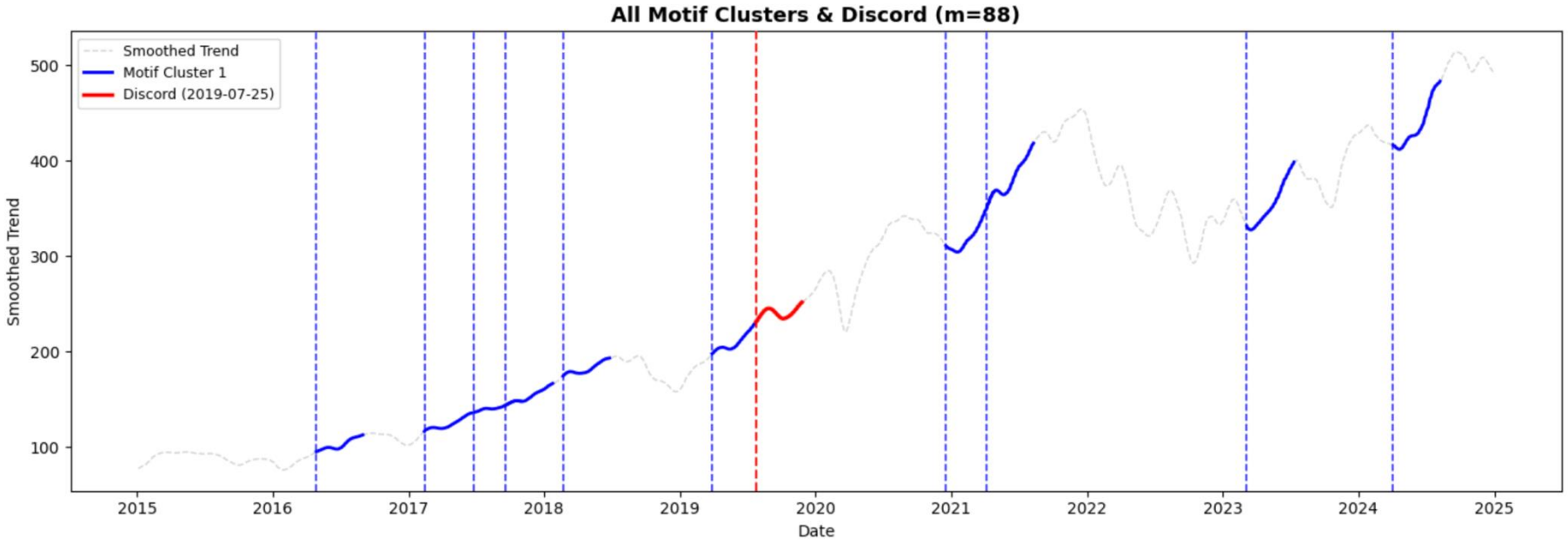
- Do the directional momentum shapes repeat

Motif Analysis - Process



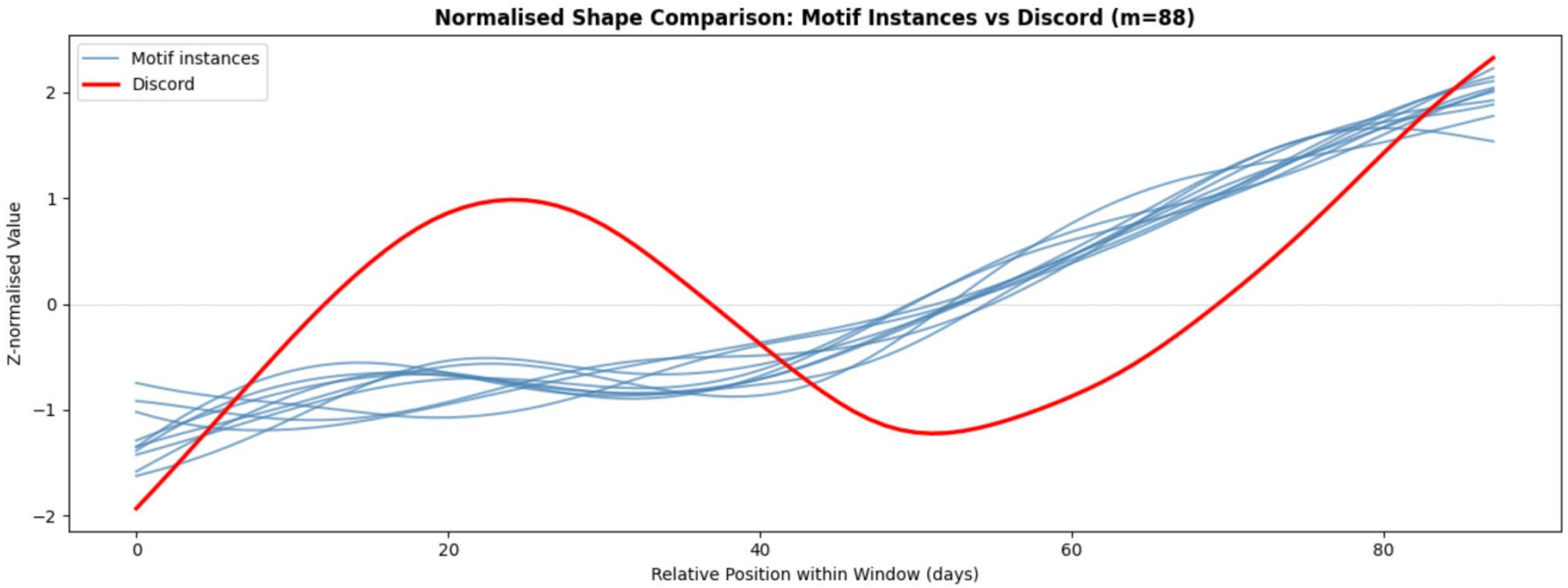
- Apply heuristic search method for a good motif window length m for Matrix Profile
- Candidates for search m_cand : [88, 176, 352, 640] – window in days
- For each sliding window of length m_cand, find its nearest non-trivial match elsewhere in the series
  - 88-day subsequences have **closer nearest neighbours** elsewhere in the series
  - **37 valleys**, more than the others - many local segments of length 88 appear to have strong recurrence
  - **Shorter windows** are finding more repeated local shapes

Motif Analysis – Find Cluster Patterns



- **One dominant motif cluster** with multiple occurrences
- **Recurrent 88-day configuration** that reappears across multiple market periods
- **Blue lines** show the **start dates of the 10 occurrences** of the dominant 88-day motif cluster
- **Red line** marks the most **anomalous 88-day subsequence ("discord")**

Motif Analysis – Pattern Trajectories (Shape)



- Normalized (remove absolute level and scale) - the internal trajectory within each 88-day window
- **Motif Shapes** - slow base / consolidation followed by an upward acceleration or trend resumption
  - Motif lines are similar enough that the motif family has a clear common shape
- **Discord Shape** - more wave-like / oscillatory / reversal-heavy structure
  - Discord a materially different internal trajectory

Motif Analysis – Query-by-Latest Period Analogue search

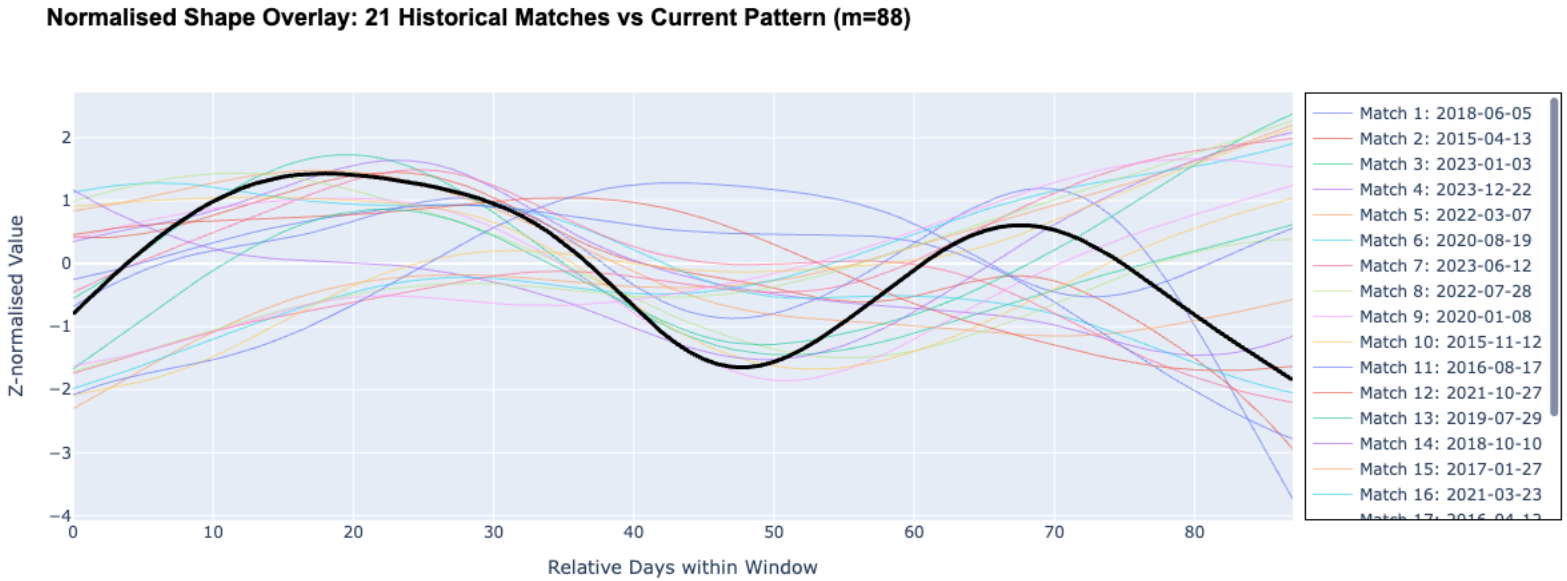
Given **most recent m-length pattern** in **smoothed_trend** searches full history for **similar past windows**

- **Which past segments of length m look most like the current last *m* observations**
- No heavy overlap – enforce minimum spacing of m between selected matches
- No clusters of nearly identical neighbouring windows
- Each selected match represents a more distinct historical episode

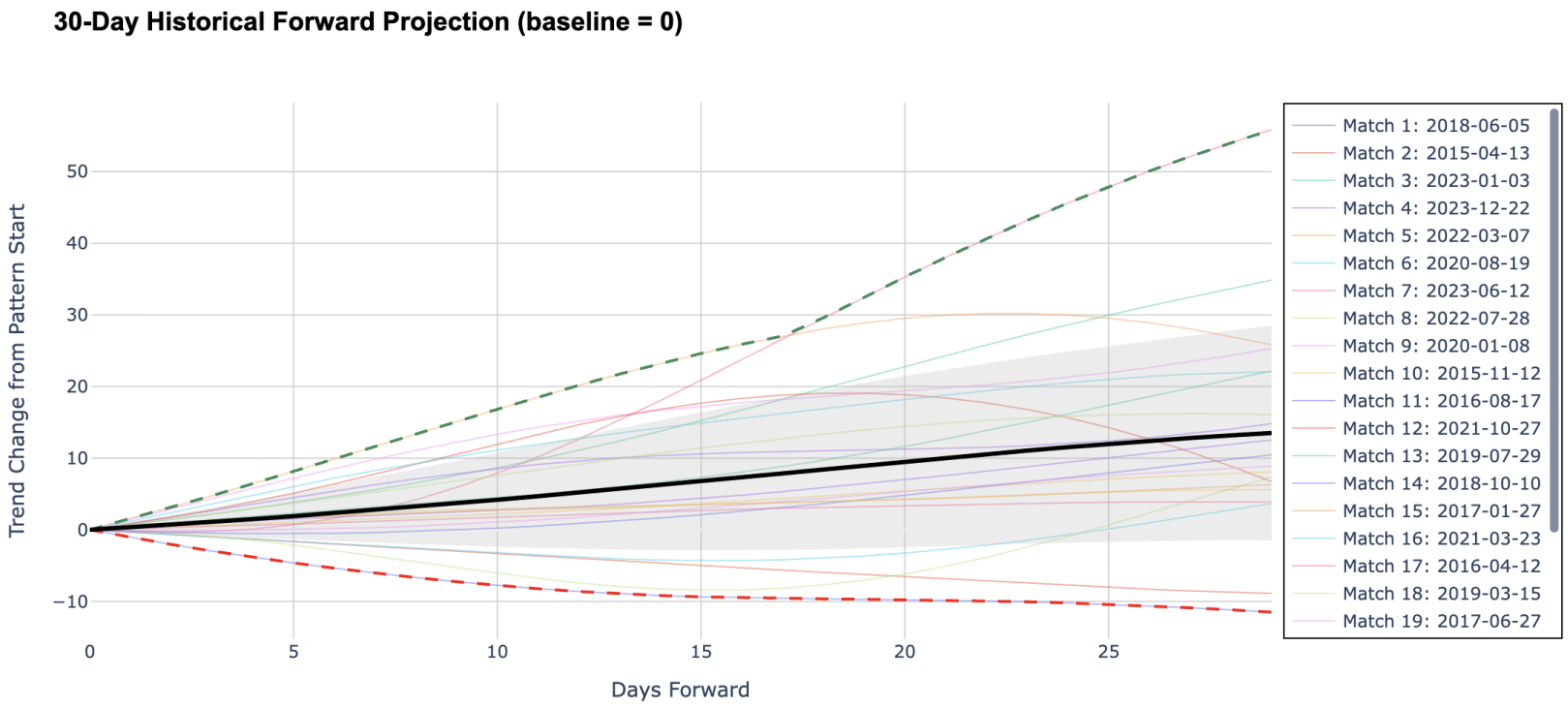
- **De Facto : Query-by-Latest Period example subsequence matching**

- **Differentiating Motif Analysis & Query-by-Latest Period**
  - Motif discovery - Search for repeated similar patterns anywhere in the series
  - Nearest-neighbour - query-by-example subsequence matching

Motif Analysis – Query-by-Latest Period Analogue search

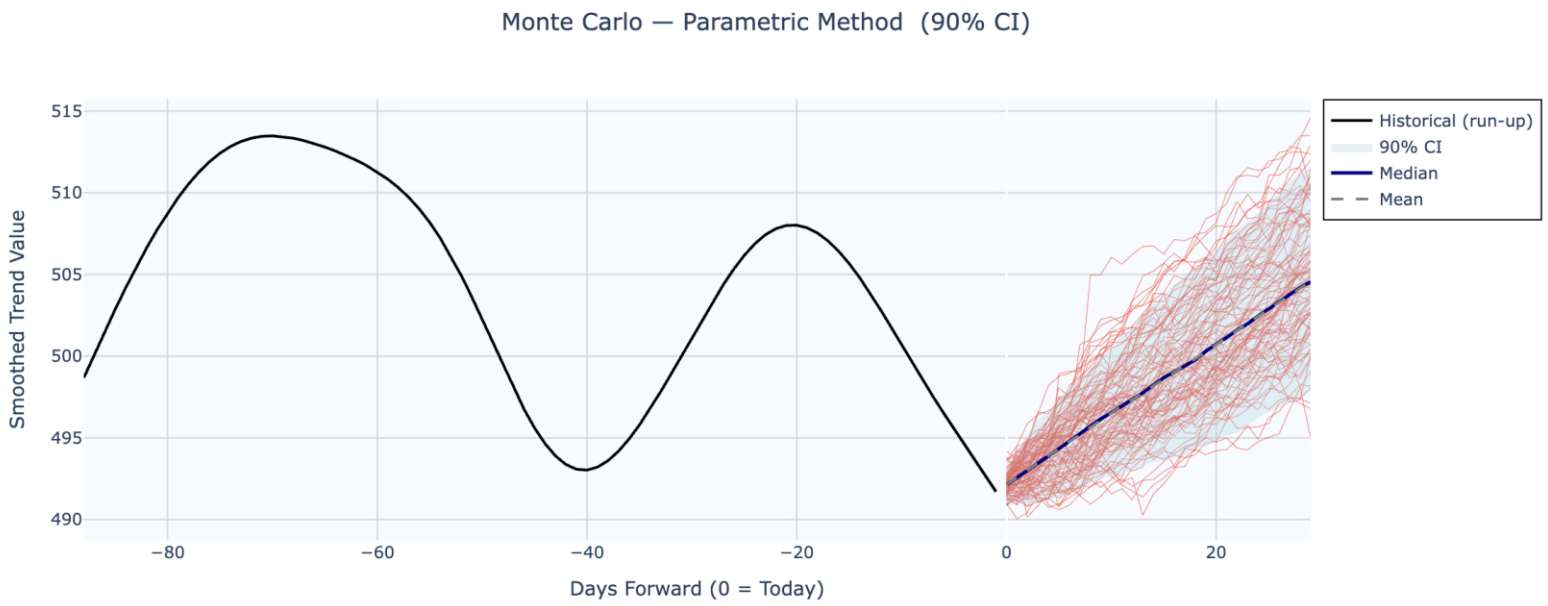


Motif Analysis – Historical Analogue Forward Projection (*t+30*)



- **Not a parametric** forecast
- **Historical analogue projection**
  1. first **identify past windows most similar to the current pattern**
  2. examine **the realized 30-day paths that followed** those historical matches
- Black line - the **average forward path**
- Grey band shows **cross-episode dispersion**

Motif Analysis – Parametric Monte Carlo Forward Projection(*t+30*)



- **Daily changes are simulated from a fitted distribution calibrated to the historical post-match change pool**
- **Fan** reflects a model-based **range of plausible future paths**
- **Median/mean lines** summarize the central projected trajectory

Motif Analysis – Summary

- **Motif analysis** detects recurring **local patterns** in time series and is relevant since financial and macro data are often **nonlinear, state-dependent, asymmetric, noisy, and regime-driven**

- The analysis focused on three distinct pattern-based ideas: **motifs** (repeated shapes), **discords** (anomalous subsequences), and **query-by-latest-period matching** (historical analogues to the current pattern)

- **The heuristic window search** selected **m = 88**, suggesting that the strongest repeated structure in this series appears at that timescale; motif extraction then identified **one dominant recurring motif family** and **one clear discord**

- **Query-by-latest-period matching** showed that the **current pattern** has **multiple meaningful historical analogues**, and the associated **forward paths** provided an **empirical view of what tended to happen next after similar episodes**

- Overall, the main learning is that **pattern-based methods complement predictive models by making recurrence, anomaly, and analogue structure explicit**, providing interpretable context for forecasting and regime analysis